Assessment Tool Co2-Tool-1

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1. The table below shows the annual premium increase rate (x_i) and the number of claims per 100 policyholders (y_i) in a health insurance company. Using the covariance formula, determine whether premium increase and claim frequency have a positive or inverse relationship. Before computing covariance, calculate the mean of $x$and y.

PREMIUM INCREASE (%) ((X_I))	CLAIMS PER 100 POLICYHOLDERS ((Y_I))
3.2	15
4.0	18
5.5	25
4.8	20

Answer

Step 1: Calculate the Mean

Mean of Premium Increase:

$$\bar{x} = \frac{3.2 + 4.0 + 5.5 + 4.8}{4} = \frac{17.5}{4} = 4.375$$

Mean of Claims:

$$\bar{y} = \frac{15 + 18 + 25 + 20}{4} = \frac{78}{4} = 19.5$$

Step 2: Calculate Covariance

Formula:

$$\text{Cov}(X, Y) = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n}$$

Calculations:

- (3.2 − 4.375) (15 − 19.5) = 5.2875
- (4.0 − 4.375) (18 − 19.5) = 0.5625
- (5.5 − 4.375) (25 − 19.5) = 6.1875
- (4.8 − 4.375) (20 − 19.5) = 0.2125

Sum of products:

$$5.2875 + 0.5625 + 6.1875 + 0.2125 = 12.25$$

Covariance:

$$\text{Cov}(X, Y) = \frac{12.25}{4} = 3.0625$$

Final Answer

- Mean of Premium Increase ($\bar{x}$) = **4.375**
- Mean of Claims ($\bar{y}$) = **19.5**
- Covariance = **3.0625**

Conclusion

Since the covariance is positive (3.0625), there is a positive relationship between premium increase and claim frequency. As the premium increase percentage rises, the number of claims per 100 policyholders also tends to increase.

2. A hospital is analyzing patient blood pressure readings (systolic) to identify trends. Due to minor fluctuations, the analyst performs local smoothing using equal-frequency (equi-depth) binning.

Sample Data Set (Systolic BP Readings):

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

Answer:

Step 1: Number of Data Values

Total values = **27**

Divide the data into **3 equal-frequency bins**.

Number of values in each bin = **9**

Step 2: Form the Bins

Bin 1:

110, 112, 115, 115, 118, 120, 120, 122, 124

Bin 2:

124, 126, 126, 126, 126, 130, 135, 135, 138

Bin 3:

138, 138, 140, 145, 150, 152, 160, 180, 138

(After sorting Bin 3)

138, 138, 138, 140, 145, 150, 152, 160, 180

Step 3: Perform Local Smoothing (Replace each value with the Bin Mean)

Bin 1 Mean

$$\frac{110 + 112 + 115 + 115 + 118 + 120 + 120 + 122 + 124}{9}$$

$$= \frac{1056}{9} = 117.33$$

Smoothed Bin 1:

117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33

Bin 2 Mean

$$\frac{124 + 126 + 126 + 126 + 126 + 130 + 135 + 135 + 138}{9}$$

$$= \frac{1166}{9} = 129.56$$

Smoothed Bin 2:

129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56

Bin 3 Mean

$$\frac{138 + 138 + 138 + 140 + 145 + 150 + 152 + 160 + 180}{9}$$

$$= \frac{1341}{9} = 149.00$$

Smoothed Bin 3:

149, 149, 149, 149, 149, 149, 149, 149, 149

Final Answer

Equal-Frequency Bins

- **Bin 1:** 110, 112, 115, 115, 118, 120, 120, 122, 124
- **Bin 2:** 124, 126, 126, 126, 126, 130, 135, 135, 138
- **Bin 3:** 138, 138, 138, 140, 145, 150, 152, 160, 180

Smoothed Data (Using Bin Mean)

- Bin 1 Mean = 117.33
- Bin 2 Mean = 129.56
- Bin 3 Mean = 149.00

Smoothed Dataset:

117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33,

129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56,

149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00

3. An insurance company evaluates Body Mass Index (BMI) of 18 clients for risk classification. To simplify analysis, they smooth the data by replacing values with the closest bin boundaries using equal-width partitioning.

Sample Data Set (Sorted BMI):

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Answer:

Step 1: Calculate the Bin Width

Minimum value = **18.5**

Maximum value = **39.5**

Range:

$$39.5 - 18.5 = 21$$

Assume **3 equal-width bins**.

Bin Width:

$$\frac{21}{3} = 7$$

Step 2: Form the Equal-Width Bins

- **Bin 1:** 18.5 – 25.5
- **Bin 2:** 25.5 – 32.5
- **Bin 3:** 32.5 – 39.5

Step 3: Replace Each Value with the Closest Bin Boundary

Bin 1 (18.5–25.5)

Original Values:

18.5, 19.2, 21.8, 24.9, 25.5

Smoothed Values:

18.5, 18.5, 18.5, 25.5, 25.5

Bin 2 (25.5–32.5)

Original Values:

26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4

Smoothed Values:

25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5

Bin 3 (32.5–39.5)

Original Values:

33.1, 33.6, 34.7, 38.2, 39.5

Smoothed Values:

32.5, 32.5, 32.5, 39.5, 39.5

Final Answer

Equal-Width Bins

- **Bin 1:** 18.5 – 25.5
- **Bin 2:** 25.5 – 32.5
- **Bin 3:** 32.5 – 39.5

Smoothed Data (Nearest Bin Boundary)

18.5, 18.5, 18.5, 25.5, 25.5,

25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5,

32.5, 32.5, 32.5, 39.5, 39.5

Conclusion

The BMI values are grouped into **three equal-width bins**, and each value is replaced with its **nearest bin boundary**, reducing minor variations and making the data easier to analyse for risk classification.

4. A hospital administration is reviewing the cost of medical consumables (in ₹). To reduce data for mining purposes, they discretize the sorted price records using bin medians.

Sample Data Set (Costs in ₹):

120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

Answer:

Step 1: Number of Data Values

Total values = **27**

Divide the data into **3 equal-frequency bins**.

Number of values in each bin = **9**

Step 2: Form the Bins

Bin 1:

120, 140, 145, 145, 148, 150, 150, 150, 155

Bin 2:

155, 155, 155, 160, 160, 165, 170, 170, 175

Bin 3:

175, 175, 175, 180, 190, 200, 210, 230, 300

Step 3: Find the Median of Each Bin

Bin 1 Median

Middle (5th) value = **148**

Bin 2 Median

Middle (5th) value = **160**

Bin 3 Median

Middle (5th) value = **190**

Step 4: Replace Each Value with the Bin Median

Bin 1:

148, 148, 148, 148, 148, 148, 148, 148, 148

Bin 2:

160, 160, 160, 160, 160, 160, 160, 160, 160

Bin 3:

190, 190, 190, 190, 190, 190, 190, 190, 190

Final Answer

Equal-Frequency Bins

- **Bin 1:** 120, 140, 145, 145, 148, 150, 150, 150, 155
- **Bin 2:** 155, 155, 155, 160, 160, 165, 170, 170, 175
- **Bin 3:** 175, 175, 175, 180, 190, 200, 210, 230, 300

Bin Medians

- **Bin 1 Median = ₹148**
- **Bin 2 Median = ₹160**
- **Bin 3 Median = ₹190**

Smoothed Data (Using Bin Medians)

148, 148, 148, 148, 148, 148, 148, 148, 148,

160, 160, 160, 160, 160, 160, 160, 160, 160, 190, 190, 190, 190, 190, 190, 190, 190, 190

Conclusion

The medical supply costs are divided into three equal-frequency bins, and each value is replaced by the median of its bin, reducing data variation while preserving the central value of each group.

5. Given the following **ages of insured individuals**:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

(a) Compute the **Mean Absolute Deviation (MAD)** of age.

(b) Compute the **z-score** for the first four observations.

Answer:

Step 1: Calculate the Mean

Mean

$$= \frac{18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28}{10}$$

$$= \frac{330}{10} = 33$$

(a) Mean Absolute Deviation (MAD)

Formula:

$$\text{MAD} = \frac{\sum | x - \bar{x} |}{n}$$

Absolute deviations from the mean (33):

- $|18 - 33| = 15$
- $|22 - 33| = 11$
- $|25 - 33| = 8$
- $|42 - 33| = 9$
- $|28 - 33| = 5$
- $|43 - 33| = 10$
- $|33 - 33| = 0$
- $|35 - 33| = 2$
- $|56 - 33| = 23$
- $|28 - 33| = 5$

Sum of absolute deviations:

$$15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5 = 88$$

Mean Absolute Deviation:

$$\text{MAD} = \frac{88}{10} = 8.8$$

Answer (a):

MAD = 8.8

(b) Compute the z-score for the First Four Observations

Formula:

$$z = \frac{x - \bar{x}}{\sigma}$$

Population Standard Deviation:

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \sqrt{\frac{1174}{10}} = \sqrt{117.4} \approx 10.84$$

Now calculate the z-scores:

- **For 18:**

$$z = \frac{18 - 33}{10.84} = \frac{-15}{10.84} \approx -1.38$$

- **For 22:**

$$z = \frac{22 - 33}{10.84} = \frac{-11}{10.84} \approx -1.02$$

- **For 25:**

$$z = \frac{25 - 33}{10.84} = \frac{-8}{10.84} \approx -0.74$$

- **For 42:**

$$z = \frac{42 - 33}{10.84} = \frac{9}{10.84} \approx 0.83$$

Final Answer

- Mean = 33
- Mean Absolute Deviation (MAD) = 8.8
- Standard Deviation ≈ 10.84

Z-Scores (First Four Observations)

Age	Z-Score
18	−1.38
22	−1.02
25	−0.74
42	0.83